

Medium Assessment – Cortex Text Translation (Practice)

Assessment Type

Medium

Duration: 25 minutes

Skill Focus

- Snowflake Cortex translation
- Handling multiple languages
- Conditional translation logic
- Output validation

Rules

- Use **ONLY SNOWFLAKE.CORTEX.TRANSLATE()**
 - Do **NOT** use COMPLETE() or SUMMARIZE()
 - Do **NOT** hardcode translated text
 - Handle NULL / empty text safely
-

1) Setup and Dataset

Dataset

SUPPORT_MESSAGES — used for multi-language translation.

Columns

- MESSAGE_ID (INT)
 - SOURCE_LANGUAGE (STRING)
 - MESSAGE_TEXT (STRING)
 - CHANNEL (STRING)
 - CREATED_AT (TIMESTAMP_NTZ)
-

2) Tasks to Perform (Step-by-Step Activities)

Task 1 – Basic Language Translation

Objective

Translate message text from **English to Tamil** using Cortex.

Steps

- Inspect the message content.
- Apply SNOWFLAKE.CORTEX.TRANSLATE() on MESSAGE_TEXT.
- Generate Tamil translation.

Output Columns

- MESSAGE_ID
- MESSAGE_TEXT
- TRANSLATED_TEXT (Tamil)

Validation

- Meaning preserved
 - Correct Tamil output
 - No extra explanation added
-

Task 2 – Reverse Translation

Objective

Translate **Tamil messages back to English**.

Steps

- Filter messages where SOURCE_LANGUAGE = 'Tamil'
- Translate text to English.

Validation

- Output language is English
 - Sentence structure is logical
-

Task 3 – Conditional Translation Based on Source Language

Objective

Translate dynamically based on SOURCE_LANGUAGE.

Rules

- If source = English → Tamil
- If source = Tamil → English

Steps

- Use conditional logic
- Apply CORTEX.TRANSLATE() accordingly

Validation

- No row is skipped
 - Correct target language per row
-

```
CREATE OR REPLACE TABLE TEXT_MESSAGES (  
  MSG_ID INT,  
  SOURCE_LANG STRING,  
  MESSAGE STRING  
);
```

```
INSERT INTO TEXT_MESSAGES VALUES  
(1, 'English', 'Delivery is delayed and needs urgent attention.'),  
(2, 'English', 'The refund has not been processed yet.'),  
(3, 'English', 'Customer support resolved the issue quickly.'),  
(4, 'Tamil', 'உங்கள் ஆர்டர் இன்று அனுப்பப்படும்.'),  
(5, 'English', 'App crashes during checkout.'),  
(6, 'English', NULL);
```

```
1 SELECT  
  MSG_ID,
```

```
MESSAGE,  
SNOWFLAKE.CORTEX.TRANSLATE(  
    MESSAGE,  
    'Tamil',  
    'English'  
) AS TRANSLATED_TEXT  
FROM TEXT_MESSAGES  
WHERE SOURCE_LANG = 'English';
```

```
2 SELECT  
    MSG_ID,  
    MESSAGE,  
    SNOWFLAKE.CORTEX.TRANSLATE(  
        MESSAGE,  
        "English",  
        'Tamil'  
    ) AS TRANSLATED_TEXT  
FROM TEXT_MESSAGES  
WHERE SOURCE_LANG = 'Tamil';
```

```
3 SELECT  
    MSG_ID,  
    MESSAGE,  
    CASE  
        WHEN SOURCE_LANG = 'English' THEN  
            SNOWFLAKE.CORTEX.TRANSLATE(MESSAGE, 'English', 'Tamil')  
        WHEN SOURCE_LANG = 'Tamil' THEN
```

```
    SNOWFLAKE.CORTEX.TRANSLATE(MESSAGE, 'Tamil', 'English')  
END AS TRANSLATED_TEXT  
FROM TEXT_MESSAGES;
```